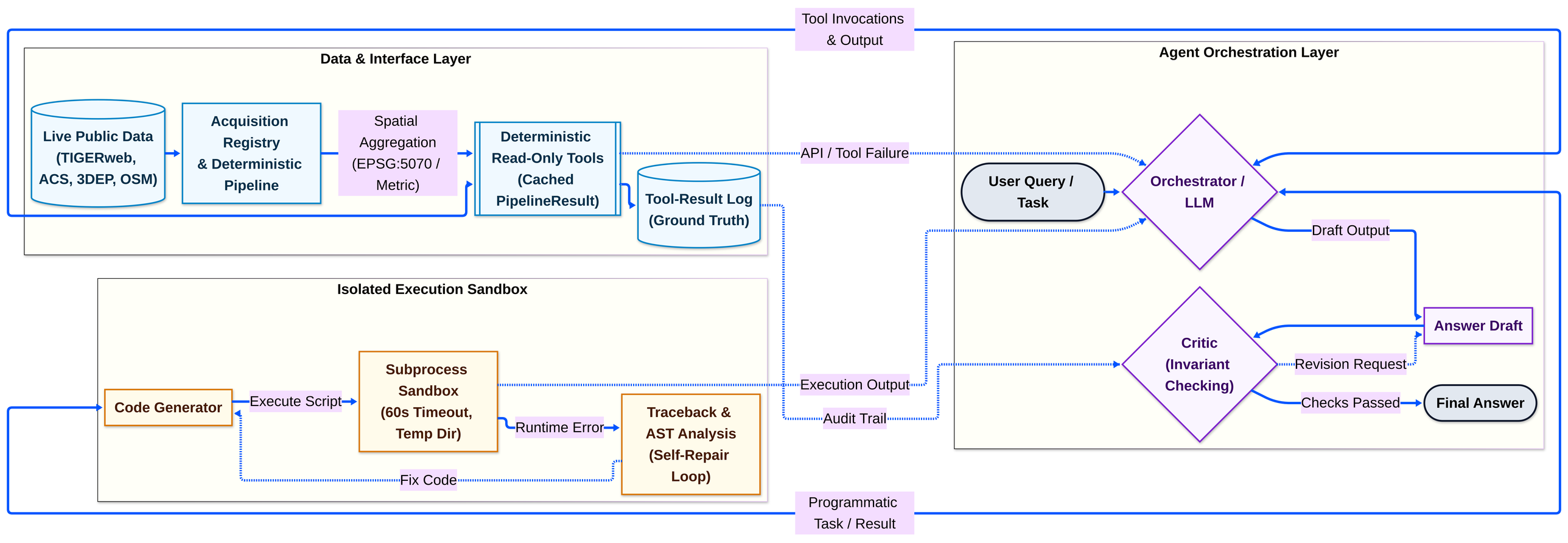

Tool Invocations & Output
Data & Interface Layer
Live Public Data (TIGERweb, ACS, 3DEP, OSM)
Acquisition Registry & Deterministic Pipeline
Spatial Aggregation (EPSG:5070 / Metric)
Deterministic Read-Only Tools (Cached PipelineResult)
Tool-Result Log (Ground Truth)
API / Tool Failure
Agent Orchestration Layer
User Query / Task
Orchestrator / LLM
Draft Output
Answer Draft
Critic (Invariant Checking)
Revision Request
Checks Passed
Final Answer
Isolated Execution Sandbox
Code Generator
Execute Script
Subprocess Sandbox (60s Timeout, Temp Dir)
Runtime Error
Traceback & AST Analysis (Self-Repair Loop)
Fix Code
Execution Output
Audit Trail
Programmatic Task / Result